

Input series: 23 ch $\times$ 256

FiLM block 1: 23 $\rightarrow$ 18 ch, $k=6$
length 256 $\rightarrow$ 251 $\rightarrow$ 125

FiLM block 2: 18 $\rightarrow$ 15 ch, $k=5$
length 125 $\rightarrow$ 121 $\rightarrow$ 60

FiLM block 3: 15 $\rightarrow$ 10 ch, $k=4$
length 60 $\rightarrow$ 57 $\rightarrow$ 28

FiLM block 4: 10 $\rightarrow$ 10 ch, $k=4$
length 28 $\rightarrow$ 25 $\rightarrow$ 12

FiLM block 5: 10 $\rightarrow$ 10 ch, $k=4$
length 12 $\rightarrow$ 9 $\rightarrow$ 4

Flatten: 10 $\times$ 4 = 40

Classifier MLP
40 $\rightarrow$ 50 $\rightarrow$ 20 $\rightarrow$ n_{classes} (LeakyReLU)

Logits (N, n_{classes})

γ, β

Extra features
or constant 1 if none

FiLM generator
Linear 32 $\rightarrow$ LReLU $\rightarrow$ 126

126 = 2 $\times$ (18+15+10+10+10)
split into $\gamma = 1 + \Delta\gamma$, β per block
zero-init $\Rightarrow$ identity at start

Inside a FiLM block
Conv1d $\rightarrow$ BatchNorm $\rightarrow$ LeakyReLU
 $\rightarrow h \leftarrow \gamma \odot h + \beta$
 $\rightarrow$ MaxPool(2)
 $\gamma, \beta \in \mathbb{R}^{N \times C}$, broadcast over time